

Winter Examination – 2022

Semester: V

Duration: 3 Hr.

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

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$$\text{a) } X(Z) = \frac{2+Z^{-1}}{(1-Z^{-1})(1+0.5Z^{-1})}$$

$$\text{b) } X(Z) = \frac{1-3Z^{-1}}{(1-0.1Z^{-1})(1+0.6Z^{-1})}$$

- C) Determine the Z-transform of given signal. Depict ROC and locations of poles and zeros in Z plane. (L3/CO4) 6

$$x(n) = \left(\frac{1}{2}\right)^n u(n) + 2^n u(n)$$

Q.4 Solve Any Two of the following. 12

- A) Differentiate between Finite Impulse Response (FIR) filter and Infinite Impulse Response (IIR) filter. (L2/CO2) 6

- B) Transform the analog filter transfer function into digital filter $H(Z)$ using Impulse Invariant Method (L3/CO1) 6

$$H_a(S) = \frac{0.5(S+4)}{(S+1)(S+2)}$$

- C) For transfer function $H(S)$ find $H(Z)$ using Bilinear transformation method. Assume $T=1$. (L2/CO2) 6

$$H(S) = \frac{1}{(S^2 + \sqrt{2}S + 1)}$$

Q. 5 Solve Any Two of the following. 12

- A) A digital Butterworth filter has to be designed using Bilinear transformation. The filter specifications are as follows: (L3/CO2) 6

$$\begin{aligned} 0.9 \leq |H(e^{j\omega})| \leq 1 & \quad 0 \leq \omega \leq 0.5\pi \\ |H(e^{j\omega})| \leq 0.2 & \quad 0.75\pi \leq \omega \leq \pi \end{aligned}$$

Find the filter order N and cut-off frequency Ω_c

- B) Find the filter order N and cut-off frequency Ω_c for an IIR low pass Butterworth filter using Bilinear transformation. The filter specifications are as follows: (L2/CO4) 6

$$\begin{aligned} \text{Passband: } 0.8 \leq |H(e^{j\omega})| \leq 1 & \quad |\omega| \leq 0.2\pi \\ \text{Stopband: } |H(e^{j\omega})| \leq 0.2 & \quad 0.6\pi \leq |\omega| \leq \pi \end{aligned}$$

Assume $T=1$ second.

- C) Consider the following LTI system with system function and draw the direct form I and direct form II structure (L2/CO2) 6

$$\text{a) } H(Z) = 1 - \frac{1}{3}Z^{-1} + \frac{1}{6}Z^{-2} + Z^{-3}$$

$$\text{b) } H(Z) = \frac{1+2Z^{-1}-Z^{-2}}{1+Z^{-1}-Z^{-2}}$$

*** End ***